



Data Science Project Technical Specification

Diarmuid O'Neill

South East Technological University

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Table of Contents

Introduction.....	3
Research Question	4
Dataset.....	5
Methods	6
References	7

Introduction

This technical specification aims to outline some research questions and methods relating to sleep health and lifestyle. It will also describe the chosen dataset and specific statistical models that will be used to conduct the investigation.

Research Question

This project aims to investigate if sleep and lifestyle factors can predict the presence of sleep disorders. Specifically which factors most significantly influence the presence of sleep disorders, what kind of sleep disorders can be predicted and what factors most affect sleep quality. It is expected that stress level, sleep duration and physical activity will be the most significant predictors of sleep disorders. It is also expected that stress will be a key contributor to lower sleep duration.

Dataset

The dataset used for this investigation is a sample Sleep Health and Lifestyle dataset. This dataset was originally published on Gigasheet and has been accessed on Kaggle. According to its publisher on Kaggle it is a structured and cleaned version of a public sample data set intended for research and educational purposes (Shamim, 2025).

The dataset is tabular and contains 374 rows and 13 columns. Variables include age, gender occupation, physical activity, stress level, daily steps, heart rate, blood pressure and more. The target variable sleep disorder indicates whether the individual has no disorder, insomnia or sleep apnoea (Shamim, 2025).

Methods

Python will be used in Jupyter Notebook to work with the dataset. Some libraries that will be needed are the scikit-learn (sklearn) for building statistical models, matplotlib for visualizing the data and pandas for general data manipulation.

Initially the dataset will be pre-processed and cleaned. After this, Exploratory Data Analysis (EDA) will be conducted to help visualize relationships between variables and to try find any obvious correlations or predictors that were not previously considered.

The dataset will be split into training and testing subsets to train and test models effectively. Gaussian naïve bayes and some tree-based models (such as Decision Trees and Random Forests) will be used to predict the probability of an individual having a sleeping disorder given certain factors such as sleep duration being less than six hours for instance. These models have been chosen as they are well suited for categorical prediction.

Linear regression will also be used to find potential correlations between sleep quality and other features such as stress and physical activity level. This method has been chosen as it more suited to continuous quantitative data and can show the strength and direction of associations.

References

Shamim, A. (2025, October 27). *Sleep Health and Lifestyle Dataset*. Retrieved from Kaggle:
<https://www.kaggle.com/datasets/adilshamim8/sleep-cycle-and-productivity>